

Engineering Reboot 2026

Week-02 Engineering Lab Workbook v3.1

Review Draft

Workbook Philosophy

This workbook is designed as a self-contained execution package. A participant should be able to complete the entire week without consulting separate manuals, governance documents, or reference material.

Day 06 – Conditional Logic

Mission Brief

Conditional logic is the foundation of software decision making. Authentication systems, industrial controls, home automation, and embedded systems rely heavily on conditional execution.

Exercise 01 – Age Classifier

Program Filename

age_classifier.py

Required Prompt

Input age:

Classification Rules

0-12 = Child

13-19 = Teen

20-64 = Adult

65+ = Senior

Required Test Cases

Test Case T01

Input age: 5

Classification: Child

Test Case T02

Input age: 13

Classification: Teen

Test Case T03

Input age: 35

Classification: Adult

Test Case T04

Input age: 70

Classification: Senior

Expected Console Transcript

```
C:\Projects\Engineering-Reboot-2026> python age_classifier.py
```

Input age: 5

Classification: Child

Screenshot Deliverables

AgeClassifier-Code.png

AgeClassifier-Test-T01.png

AgeClassifier-Test-T02.png

AgeClassifier-Test-T03.png

AgeClassifier-Test-T04.png

Acceptance Criteria

Program executes successfully.

All four test cases pass.

Screenshots captured.

Git commit completed.

Git push completed.

Day 07 – Boolean Logic

Mission Brief

Focused practice on boolean logic using executable exercises, evidence collection, and Git workflow.

Program Filename

eligibility_checker.py

Workbook Structure

Objective

Input Specification

Expected Display Format

Required Test Matrix

Screenshot Requirements

Acceptance Criteria

Git Commit Procedure

Review Note

Final locked edition will expand every exercise with complete prompts, exact outputs, explicit test matrices, screenshot filenames, and command blocks.

Day 08 – Loops

Mission Brief

Focused practice on loops using executable exercises, evidence collection, and Git workflow.

Program Filename

counting_program.py

Workbook Structure

Objective

Input Specification

Expected Display Format

Required Test Matrix

Screenshot Requirements

Acceptance Criteria

Git Commit Procedure

Review Note

Final locked edition will expand every exercise with complete prompts, exact outputs, explicit test matrices, screenshot filenames, and command blocks.

Day 09 – Loop Practice

Mission Brief

Focused practice on loop practice using executable exercises, evidence collection, and Git workflow.

Program Filename

pattern_generator.py

Workbook Structure

Objective

Input Specification

Expected Display Format

Required Test Matrix

Screenshot Requirements

Acceptance Criteria

Git Commit Procedure

Review Note

Final locked edition will expand every exercise with complete prompts, exact outputs, explicit test matrices, screenshot filenames, and command blocks.

Day 10 – Mini Project

Mission Brief

Focused practice on mini project using executable exercises, evidence collection, and Git workflow.

Program Filename

menu_driven_calculator.py

Workbook Structure

Objective

Input Specification

Expected Display Format

Required Test Matrix

Screenshot Requirements

Acceptance Criteria

Git Commit Procedure

Review Note

Final locked edition will expand every exercise with complete prompts, exact outputs, explicit test matrices, screenshot filenames, and command blocks.